

Week 4 Assignment: 3 The Good Casino

1 Objective

The goal of this assignment is to simulate a complex, dynamic environment where game probabilities change every round. You will evaluate how behavioral heuristics compare against a dynamically updating, mathematically optimized Quant strategy.

2 The Dynamic Game Environment

In a real life casino, the house (or casino) always has an edge, i.e. in the long run *the casino always wins*. But suppose you are a card counter and playing Black Jack. By card counting you may sometime have a positive edge too, depending on the shoe deck remaining. Three players are going to play with different mindsets. Let us observe who wins in long run. The winning probability changes at each step t .

2.1 Environmental Parameters

- **Base Win Probability (p_0):** 0.50 (Even odds).
- **Dynamic Market Edge (α_t):** Every round t , a random variable α_t is drawn from a uniform distribution between -0.05 and $+0.07$. This represents a shifting advantage (e.g., shoe deck count or temporary market alpha).
- **Round Win Probability (p_t):** Calculated dynamically each round as $p_t = p_0 + \alpha_t$.
- **Loss Probability (q_t):** Calculated as $q_t = 1 - p_t$.
- **Payout Metric (b):** 1 : 1 (Returns \$1 net profit for every \$1 risked).
- **Starting Capital (W_0):** \$1,000 for all concurrent players.
- **Simulation Depth (T):** 1,000 rounds.

3 Task: Implementation of Strategies

You must develop a Python script where all three players run simultaneously through the exact same sequence of environmental shifts (α_t) and game outcomes.

3.1 Player 1: The Random Gambler

- **Execution:** Every round t , this player ignores the changing environment (α_t) and randomly selects a uniform integer percentage $k \in [1, 10]$.

$$Bet_t = \lfloor W_t \times (k/100) \rfloor$$

3.2 Player 2: The Martingale Gambler

- **Execution:** This player ignores the environment and focuses purely on historical loss recovery.
 - Base minimum bet size is \$10.
 - Following a winning round, reset the next bet size to \$10.
 - Following a losing round, set the next bet size to $2 \times \text{Bet}_{t-1}$.
 - *Constraint:* If bankroll $W_t < \text{Bet}_t$, the player goes “all-in”.

3.3 Player 3: The Smart Gambler

- **Logic:** This player observes the environmental state (α_t) at the start of each round. Because the expected value dynamically swings between positive and negative, the Smart Gambler recalculates the optimal bet size dynamically:

$$f_t^* = \frac{b \cdot p_t - q_t}{b}$$

- **Execution:** Given $b = 1$, the equation simplifies to $f_t^* = p_t - q_t$. The Smart Gambler executes the following conditional logic:
 - **If $f_t^* > 0$ (+EV environment):** The Smart Gambler bets exactly their optimal wealth fraction: $\text{Bet}_t = f_t^* \times W_t$.
 - **If $f_t^* \leq 0$ (−EV environment):** The Smart Gambler recognizes the house has the edge and dynamically adjusts their bet to **\$0**, sitting out the round entirely.

4 Deliverable

1. **PDF/.py/.ipynb** Submit a pdf(preferably or .py/.ipynb) file mentioning the code (well commented), the charts of their timelines of bankrolls using log scale, over 1000 rounds for 100 players of each mindset, and the research part paragraph given below.
2. **Research:** Research upon this, card counting strategy in Black Jack, and its basic strategy and briefly explain in under one paragraph, why card counting beats basic strategy. And how is Black Jack the only beatable game in Casino.